Errata

Coppock, A. (2019). Generalizing from Survey Experiments Conducted on Mechanical Turk: A Replication Approach. *Political Science Research and Methods* 7(3), 613-628.

The maintained replication code behind these errata is at https://github.com/active-maintenance-aec/coppock_2019a and is archived at Zenodo at https://doi.org/10.5281/zenodo.24168306, a DOI that always resolves to the latest version.

Five sentences in the published article need correcting. Two state numbers the deposited replication archive does not support, and a third describes the tests' statistical power more generously than the corrected numbers allow. The remaining two are in the footnote that explains the Holm correction: one calls the correction a correlation, and the other states the Benjamini-Hochberg threshold on the false discovery rate in place of the Holm threshold the footnote names and the code applied. All five are in the section reporting the randomization inference tests of treatment effect homogeneity. It also carries three entries of a second kind, which quote a passage and propose no change to it: the Original and TESS/GfK cells of Table 2's N Significant column, which are interchanged; the Original and Mechanical Turk counts in the sentence that reports the table, which differ from the deposit by one each; and the sentence limiting each study to two dependent variables, which is true or false according to what counts as a study. **No correction and no note here changes a conclusion of the paper.** The tests still reject the null of constant effects in a small minority of comparisons, the counts in Table 2 come from the code rather than from the footnote's formula, and the correspondence between the Mechanical Turk and probability sample estimates is untouched.

Article DOI: 10.1017/psrm.2018.10. Replication archive: 10.7910/DVN/F1CFFM. The code behind every corrected value below is at https://github.com/active-maintenance-aec/coppock_2019a, and every one of them is recomputed when this document is rendered rather than typed into it.

Corrections are marked in **bold**. Numbers the article hedges with an approximation sign keep the article's own precision, because the approximation is the claim; fuller values appear in the note beneath each entry.

1. Simulated power, page 622

Published. "The MTurk versions of the experiments studied here typically employ 500 subjects per treatment arm, suggesting that we would be well powered (power $\approx$ **0.8**) to detect treatment

effect heterogeneity on the scale of 0.2 SD, and moderately powered for 0.1 SD (power ≈ **0.6**)."

Corrected. "The MTurk versions of the experiments studied here typically employ 500 subjects per treatment arm, suggesting that we would be **moderately** well powered (power ≈ **0.6**) to detect treatment effect heterogeneity on the scale of 0.2 SD, and **poorly** powered for 0.1 SD (power ≈ **0.15**)."

The simulation results the article deposits, and which Figure 4 is drawn from, give a rejection probability of 0.61 at 0.2 SD and 0.15 at 0.1 SD, both at 500 subjects per treatment arm. Figure 4 draws guide lines at 500 subjects and at power 0.8 and annotates the 0.2 SD curve where it crosses the second of them, which the simulation puts at 600 subjects per arm rather than 500; the sentence reads the height of the guide line rather than the curve. At those rejection probabilities the test is moderately well powered at 0.2 SD rather than well powered, and poorly powered at 0.1 SD rather than moderately so, and both descriptions are corrected with the numbers. The figure itself is correct and is not reprinted here.

2. Significant heterogeneity tests on the TESS/GfK sample, page 622

Published. "On the TESS/GfK replications, **0** of 10 tests were significant."

Corrected. "On the TESS/GfK replications, **1** of 10 tests were significant."

The deposited output file for Table 2 records 1 significant TESS/GfK test, and so does the maintained pipeline; the count does not move with the change R 3.6.0 made to `sample()`*, which moves the other two rows of that column. Table 2 prints 2 in the TESS/GfK row of its N Significant column and 1 in the Original row, which is that column read in the row order of the deposited output file rather than in the table's own; the two entries are interchanged. Entry 6 sets out the transposition, and because correcting the table requires the pre-R-3.6.0 sampler it is carried there as a note rather than reprinted corrected. The corrected count is what makes the next sentence true: it says that only Hiscox (2006) returned a significant test in more than one sample, which the printed 0 rules out, because the Original sample's single significant test is a different study. The one TESS/GfK rejection is Hiscox's negative treatment ($p = 0.01$), which also rejects on MTurk ($p < 0.01$).*

3. The condition stated for the Holm correction, footnote 8, page 623

Published. "Identify the smallest p-value for which the following condition holds: $p_k \leq \frac{k}{m}\alpha$, where k indexes the p-values, and α the target significance level."

Corrected. "Identify the smallest p-value for which the following condition holds: $p_k > \alpha/(m-k+1)$, where k indexes the p-values, and α the target significance level."

The condition the note prints is the Benjamini-Hochberg threshold on the false discovery rate, not the Holm threshold on the family-wise error rate that the note names and that the analysis used. The rest of the note is already Holm's step-down rule: it says to find the SMALLEST p-value meeting the condition and to treat it and every larger p-value as insignificant, which is where Holm stops and is not how Benjamini-Hochberg is read. So the single defect is the

condition, and correcting it makes the note describe the procedure the code ran. Both scripts call `p.adjust(max_pval, "holm")`*, the deposit's own* `coppock_generalizability_ri_analysis.R` *and the maintained* `table_2_ri_heterogeneity.R`*, so Table 2's N Significant (Holm) column and every number under it are unaffected.*

4. The description of statistical power, page 623

Published. "Because we are **relatively** well powered to detect politically meaningful differences in treatment response, I conclude from this evidence that the treatment effect homogeneity explanation of the correspondence across experimental sites is plausible."

Corrected. "Because we are **moderately** well powered to detect politically meaningful differences in treatment response, I conclude from this evidence that the treatment effect homogeneity explanation of the correspondence across experimental sites is plausible."

The sentence rests on the power figures of entry 1. At the corrected rejection probabilities, 0.61 at 0.2 SD and 0.15 at 0.1 SD, the tests are moderately well powered rather than relatively well powered, which is the description entry 1 now gives them. The conclusion the sentence draws survives on narrower evidence: with 40 MTurk tests at 500 subjects per arm, heterogeneity of 0.2 SD in every treatment would produce about 24 rejections against the 9 observed, but heterogeneity of 0.1 SD would produce about 6, so the tests rule out the first and not the second.

5. The word naming the procedure in footnote 8, page 623

Published. "To implement the **correlation**, order the m uncorrected p-values within a"family" from smallest to largest."

Corrected. "To implement the **correction**, order the m uncorrected p-values within a"family" from smallest to largest."

The note is about the Holm correction, which its own first sentence names twice, and the sentence that says how to apply it calls it a correlation. Nothing in the analysis turns on the word.

Carried, and not corrected

The three entries below are different in kind from the five corrections in this note. Each quotes a passage this edition reprints exactly as the paper published it, and proposes no change to it.

Two of them concern the counts of significant homogeneity tests, which come back only under a sampler R retired in 3.6.0, so that any corrected figure would be one the maintained pipeline cannot regenerate, or which turn on a choice of inequality that is the author's to make. The third is a sentence about the study's design that is true under one count of studies and false under another. Correcting any of the three would mean either printing a number this edition cannot reproduce or writing the author's prose for him.

Each is recorded here because an edition that silently reproduced them would leave a reader unable to tell a faithful reprint from an uncorrected error.

6. The N Significant column of Table 2, page 622

Published. "Original, N Comparisons 40, N Significant **1**, N Significant (Holm) 0

Mechanical Turk, N Comparisons 40, N Significant 9, N Significant (Holm) 8

TESS/GfK, N Comparisons 10, N Significant **2**, N Significant (Holm) 0"

Table 2 sets its rows in the order Original, Mechanical Turk, TESS/GfK, with 40, 40, and 10 comparisons, and it prints 1, 9, and 2 significant tests in those rows. The deposited output file for the table prints its rows alphabetically by sample, TESS/GfK first and original last, and its N Significant column reads 1, 9, 2 from top to bottom, which is the order the published column reads from top to bottom. The N Comparisons column was reordered to the table's own row order and the N Significant column was not, so the Original and TESS/GfK cells of that column are interchanged: in the published row order the deposit gives 2, 9, and 1. The Holm column, 0, 8, and 0, is symmetric under the interchange and so cannot show it. No corrected cell is set, because the deposit's counts come back only under the sampler R retired in 3.6.0: the maintained pipeline, on current R, returns 1, 8, and 1 in the published row order, so a corrected table would print counts this edition cannot regenerate. The TESS/GfK count of 1 does not depend on the sampler, and entry 2 corrects the sentence of the text that misreports it. No conclusion of the paper changes: the Holm column, which carries the argument, is untouched by the interchange, and the tests still reject the null of constant effects in a small minority of comparisons.

7. The Original and Mechanical Turk counts in the sentence reporting Table 2, page 622

Published. "Among the original 12 studies, just **1** of the 40 treatment versus control comparisons revealed evidence of effect heterogeneity. Among the MTurk replications, **8** of 40 treatments were shown to have heterogeneous effects."

The sentence gives 1 and 8 where the deposited output gives 2 and 9, and where Table 2 prints 1 and 9 in the Original and Mechanical Turk rows (the first of those belongs to the TESS/GfK row, as entry 6 sets out). On the deposit's own draw exactly one original test and exactly one MTurk test have a maximum p-value of exactly 0.05, so a strict inequality at 0.05 gives 1 and 8, which are the sentence's figures, and the deposit's own rule, which is not strict, gives 2 and 9. The text and the table can therefore differ legitimately by one in each of those two samples. No corrected figure is set, because correcting either would decide which rule the sentence meant, which only the author can say. The sentence's third figure, for the TESS/GfK replications, has no such account and is corrected in entry 2. No conclusion changes: under either rule a small minority of the comparisons reject the null of constant effects, which is all the sentence is used for.

8. The limit of two dependent variables per study, page 618

Published. "I limited the number of dependent variables analyzed in each study to **two**."

The deposit's analysis groups the data into 12 studies, and the largest of them carries 5 dependent variables: Johnston and Ballard (`expert_economists`) carries 5 and Hopkins and Mummolo (`frame_breadth`) 4. Under the online appendix's numbering, which splits the same data into fifteen studies, none carries more than two. The sentence is therefore false under one count of studies and true under the other, and it does not say which it means. No corrected sentence is set, because correcting it would mean choosing a count of studies and writing the author's account of his own design for him. No conclusion of the paper changes: the analysis is the same under either count, and only the description of it is at issue.